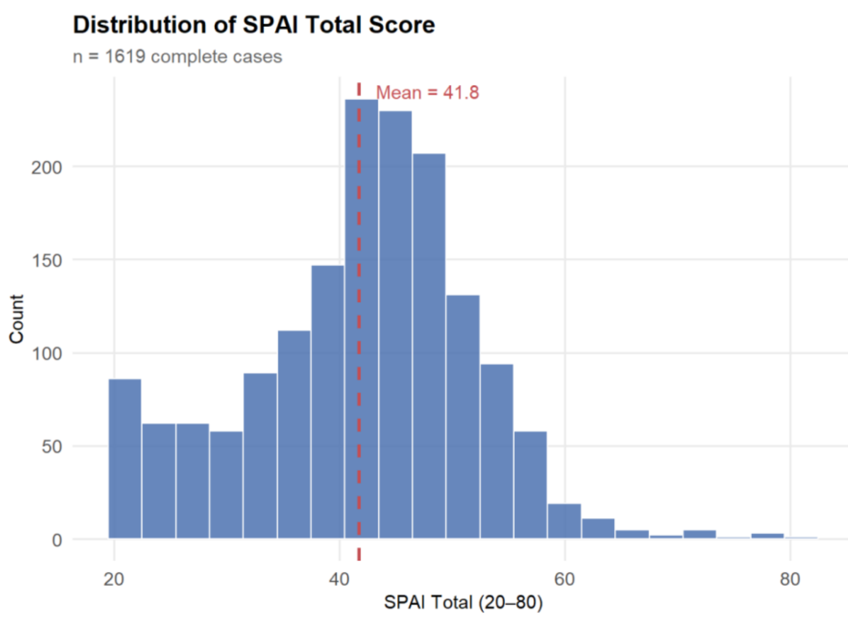
Data Visualization

Prior to model fitting, we examined the distribution of the outcome variable and the bivariate relationships between each predictor scale and SPAI total to assess the appropriateness of linear modeling assumptions and to contextualize subsequent LASSO results.

Distribution

Figure 1 displays the distribution of SPAI total scores across the 1,619 complete cases. Scores ranged from 20 to 80, with a mean of 41.8. The distribution was approximately normal with slight positive skew, suggesting that linear regression methods including LASSO are appropriate for modeling this outcome without transformation.

Figure 1: Distribution of SPAI Total Score



Correlation

Figure 2 presents Pearson correlations between each of the five predictor scale totals and SPAI total, ordered by effect size. All five scales correlated positively and significantly with SPAI total, all r > 0.55 with SAS having the highest correlation and SAPS and MPATS both having the second highest individual correlations with r = 0.71. The uniformly positive and moderate-to-strong correlations across scales suggest meaningful conceptual overlap, which is consistent with the shared nomological space of problematic smartphone use and provides motivation for a regularization approach capable of handling multicollinearity.

Figure 2: Pearson Correlations

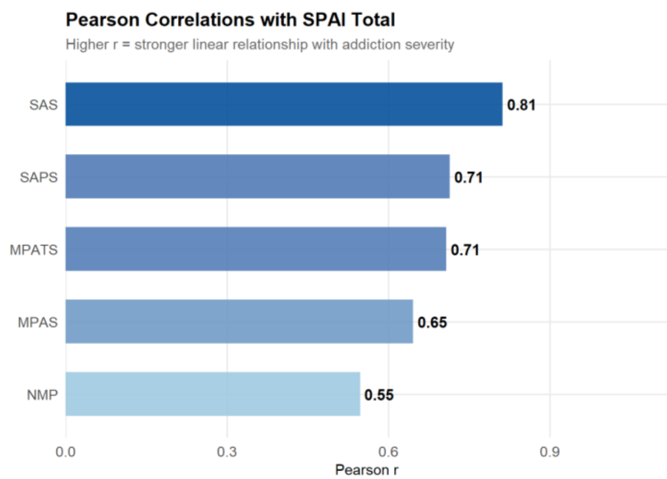


Figure 3 illustrates the bivariate relationship between SAS total and SPAI total, the strongest predictor-outcome pair identified in Figure 2. Each point represents one participant (n = 1,619). A linear trend line with 95% confidence interval is overlaid, confirming a consistent positive relationship across the full score range. The narrow confidence band and absence of notable heteroscedasticity support the assumption of homogeneous variance underlying the Gaussian LASSO models.

Figure 3: Scatter Plot (SAS vs SPAI)

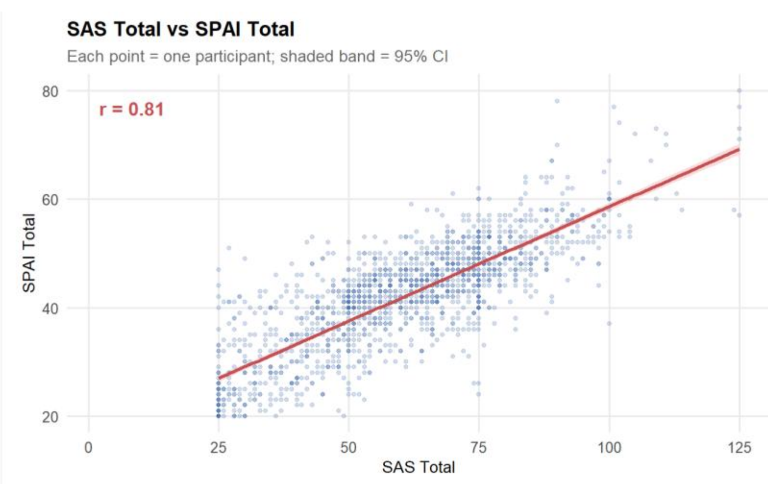


Figure 4 presents a heatmap of correlations among all six scale totals, including SPAI total. Correlations among predictor scales ranged from *r* = 0.61 to *r* = 1, indicating substantial multicollinearity. Under ordinary least squares regression, such collinearity inflates standard errors and renders individual coefficient estimates unstable. LASSO regularization addresses this directly by shrinking redundant coefficients toward zero, producing a sparse and interpretable solution even when predictors are highly intercorrelated. The pattern observed here, where all scales load on a common dimension of problematic phone use, further supports the use of a data-driven selection method rather than a priori variable exclusion.

Figure 4: Total Correlation Visualization